

BEMIDJI, MN, USAWMO: **727550**Lat: **47.500N**Lon: **94.933W**Elev: **424**StdP: **96.33**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **14958****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-27.5	-33.3	0.2	-28.0	-31.7	0.2	-25.6	11.7	-10.2	10.5	-10.1	2.4	260	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	30.0	21.0	28.0	19.7	27.1	18.9	22.6	27.9	21.3	26.4	20.2	25.0	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	16.4	26.1	19.4	14.9	24.1	18.2	13.8	23.1	68.6	27.8	63.8	26.1	60.0	25.2	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.2	8.3	-34.8	33.0	3.2	1.7	-37.1	34.3	-39.0	35.3	-40.8	36.2	-43.2	37.5		
(5)			-33.9	24.6	2.9	1.2	-35.9	25.5	-37.6	26.3	-39.2	27.0	-41.3	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-13.4	-11.1	-3.6	4.2	11.4	17.1	19.8	18.5	13.8	6.4	-2.3	-9.9		
	DBStd	12.95	8.04	7.66	7.31	5.65	4.81	3.83	3.23	3.37	4.70	5.02	6.38	7.34		
	HDD10.0	3106	726	592	423	186	41	2	0	0	16	135	369	616		
	HDD18.3	5274	984	825	681	424	221	66	22	40	148	371	618	874		
	CDD10.0	1031	0	0	1	12	83	214	303	262	130	24	1	0		
	CDD18.3	158	0	0	0	0	5	29	67	44	12	1	0	0		
	CDH23.3	1507	0	0	0	6	83	280	610	392	124	12	0	0		
	CDH26.7	357	0	0	0	1	17	66	156	91	24	2	0	0		
(14) Wind	WSAvg	3.8	3.7	3.7	3.8	4.1	4.1	3.7	3.4	3.4	3.7	4.1	3.9	3.7		
(15) Precipitation	PrecAvg	684	19	16	27	45	77	104	106	79	81	73	34	23		
	PrecMax	884	55	30	71	98	161	182	197	154	187	130	95	46		
	PrecMin	505	2	2	7	15	29	24	40	27	32	5	0	5		
	PrecStd	88	13	8	16	22	34	41	42	37	36	35	24	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.0	7.6	17.2	23.0	28.9	31.7	32.4	32.1	29.1	25.0	16.2	6.3		
		MCWB	2.1	3.5	10.4	13.0	17.8	21.8	22.9	22.3	20.3	14.6	10.4	2.3		
	2%	DB	1.4	4.1	12.3	19.2	25.9	28.5	30.0	28.9	27.0	20.5	12.3	3.1		
		MCWB	-0.2	1.8	7.4	10.3	15.6	19.8	21.5	20.6	18.8	13.6	8.1	0.9		
	5%	DB	-0.2	2.3	8.6	17.2	22.8	26.4	28.0	27.3	24.2	17.3	8.8	1.3		
		MCWB	-1.1	0.4	4.8	8.9	13.9	17.9	20.2	19.1	17.4	11.8	5.6	-0.2		
	10%	DB	-2.3	0.1	6.0	13.9	20.4	24.7	27.1	25.8	22.3	14.1	6.4	-0.3		
		MCWB	-3.4	-1.1	2.8	7.3	12.8	17.1	19.5	18.5	16.4	9.7	3.5	-1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	4.2	11.7	14.1	19.7	23.1	24.4	23.6	22.0	16.8	11.6	3.4		
		MCDB	5.1	6.8	14.3	21.5	25.7	29.4	29.7	29.4	27.1	21.9	14.1	5.3		
	2%	WB	0.2	2.2	7.7	11.2	17.2	21.2	22.9	21.9	20.2	14.5	8.2	1.3		
		MCDB	1.3	4.4	11.3	17.2	22.7	26.9	28.1	27.1	25.1	19.3	11.3	3.2		
	5%	WB	-1.3	0.4	5.1	9.6	15.4	19.7	21.6	20.6	18.6	12.3	5.8	0.0		
		MCDB	-0.4	2.2	8.3	16.3	20.5	24.6	26.7	25.1	22.9	16.5	8.7	1.3		
	10%	WB	-3.2	-1.2	3.0	7.8	13.8	18.4	20.4	19.5	17.2	10.2	3.8	-1.4		
		MCDB	-2.2	0.1	5.6	13.4	19.0	22.9	25.3	23.8	20.9	13.7	6.1	-0.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	10.9	10.6	11.3	12.0	11.1	11.4	11.5	11.1	9.1	8.1	8.2		
		MCDBR	9.7	11.4	13.5	16.3	16.3	13.5	13.1	13.3	13.9	13.9	12.2	9.5		
	5% WB	MCWBR	8.5	9.0	9.0	9.3	8.4	6.9	6.5	6.4	7.6	8.2	8.1	7.7		
		MCWBR	9.3	10.6	12.6	14.8	13.2	11.3	11.7	11.4	11.5	12.5	11.0	8.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.266	0.300	0.301	0.348	0.367	0.374	0.378	0.378	0.351	0.306	0.279	0.262		
		taud	2.319	2.214	2.391	2.360	2.367	2.373	2.381	2.401	2.455	2.566	2.516	2.390		
	Ebn at Noon	Ebn at Noon	821	855	925	905	897	889	880	865	859	858	815	788		
		Edh at Noon	72	100	99	114	117	117	115	108	93	71	60	60		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.56	2.72	3.85	4.82	5.34	5.80	6.20	5.22	3.82	2.33	1.39	1.10		
		RadStd	0.16	0.25	0.40	0.44	0.33	0.41	0.30	0.40	0.28	0.37	0.14	0.13		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (35 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page